# KIRAN ACADEMY

TASK 01: SHOPPING BILL — GST FIRST, DISCOUNT AFTER

## Solved Answer — Java and Python

This solution follows the task requirements: the five amounts are passed directly as method/function arguments; there is no Scanner, input(), or other input-taking code; each calculation method returns its amount.

### 1. Calculation and Expected Result

| Calculation | Formula | Amount |
|---|---|---|
| Electronics | (100 + 100) × 1.18 | ■236.00 |
| Fashion | (100 + 100) × 1.13 | ■226.00 |
| Food | 100 × 1.07 | ■107.00 |
| **Total after GST** | **236 + 226 + 107** | **■569.00** |
| 20% discount | 569 × 0.20 | ■113.80 |
| **Final payable amount** | **569 – 113.80** | **■455.20** |

### 2. Java Solution

```
public class ShoppingBill {

    public static double calculateElectronicsGST(double a, double b) {
        double subtotal = a + b;
        return subtotal + (subtotal * 0.18);
    }

    public static double calculateFashionGST(double c, double d) {
        double subtotal = c + d;
        return subtotal + (subtotal * 0.13);
    }

    public static double calculateFoodGST(double e) {
        return e + (e * 0.07);
    }

    public static double totalBill(double a, double b, double c,
                                   double d, double e) {
        return calculateElectronicsGST(a, b)
             + calculateFashionGST(c, d)
             + calculateFoodGST(e);
    }

    public static double calculateDiscountedBill(double a, double b,
                                                 double c, double d,
                                                 double e) {
        double total = totalBill(a, b, c, d, e);
        double discount = total * 0.20;
        return total - discount;
    }

    public static void main(String[] args) {
        double total = totalBill(100, 100, 100, 100, 100);
        double discount = total * 0.20;
        double payable = calculateDiscountedBill(100, 100, 100, 100, 100);

        System.out.printf("Total after GST = %.2f%n", total);
        System.out.printf("Discount = %.2f%n", discount);
        System.out.printf("Final payable amount = %.2f%n", payable);
    }
}
```

## 3. Python Solution

```
def calculateElectronicsGST(a, b):
    subtotal = a + b
    return subtotal + (subtotal * 0.18)


def calculateFashionGST(c, d):
    subtotal = c + d
    return subtotal + (subtotal * 0.13)


def calculateFoodGST(e):
    return e + (e * 0.07)


def totalBill(a, b, c, d, e):
    return (
        calculateElectronicsGST(a, b)
        + calculateFashionGST(c, d)
        + calculateFoodGST(e)
    )


def calculateDiscountedBill(a, b, c, d, e):
    total = totalBill(a, b, c, d, e)
    discount = total * 0.20
    return total - discount


if __name__ == "__main__":
    total = totalBill(100, 100, 100, 100, 100)
    discount = total * 0.20
    payable = calculateDiscountedBill(100, 100, 100, 100, 100)

    print(f"Total after GST = {total:.2f}")
    print(f"Discount = {discount:.2f}")
    print(f"Final payable amount = {payable:.2f}")
```

## 4. Expected Output

```
Total after GST = 569.00
Discount = 113.80
Final payable amount = 455.20
```

## 5. How the Program Works

**Step 1:** Add electronics amounts a and b, then add 18% GST.

**Step 2:** Add fashion amounts c and d, then add 13% GST.

**Step 3:** Add 7% GST to food amount e.

**Step 4:** totalBill() calls all three GST methods/functions and adds their returned amounts.

**Step 5:** calculateDiscountedBill() calculates a 20% discount on the GST-inclusive total and returns the final payable amount.

## 6. Submission Checklist

• Save Java code as **ShoppingBill.java**.

• Save Python code as **shopping_bill.py**.

• Upload both files to your GitHub repository and include the repository URL in the email.

• Send to: **9011@thekiranacademy.com**

• Subject: **Task 01 – Shopping Bill – 14/10/2026**

• Email body: Hello, I am [Your Full Name]. I have completed Task 01 – Shopping Bill in Java and Python. GitHub Repository: [Paste repository URL]